

REQUIREMENT ANALYSIS

Farm Management System - Salesforce Project

Department of Computer Application

Salesforce Project

FUNCTIONAL REQUIREMENTS

Farm Management System • Salesforce Project

Admin Functions

1. **Create Lightning App named** Farm Management System
- Create Custom **Object Tabs** for Farmer and Crop
- Manage Navigation and **Branding**
- Create Reports and Dashboards

User Functions

1. Access App from App Launcher
2. Create and Edit Farmer Records
3. Create Crop linked to Farmer
4. Update Crop Status — Planted • Growing • Harvested
5. View Reports

TECHNICAL ARCHITECTURE

Salesforce Platform:

- Lightning Experience

Custom Objects:

- Farmer__c and Crop__c with Lookup Relationship

Fields:

- Text, Phone, Email, Number
- Date, Picklist

Tabs:

- Custom Object Tabs

App:

- Lightning App: Farm Management System
- App Manager

Automation:

- Validation Rules
- Flow for email

Security:

- Profiles, Roles, OAM

Reports & Dashboards

- Analytics & Insights for Farm Performance

Farm Management System

Salesforce Project • User Roles & Permissions Overview

USER ROLES & PERMISSIONS

Role-based access structure for secure and scalable farm management on Salesforce

Full Access

1. System Administrator

- Full access to the organization
- Create Objects, Tabs, & Apps
- Manage Profiles & Roles
- Manage Users & Permissions
- Create & Manage Reports & Dashboards

Responsible: System Setup & Governance

2. Farm Manager Operational

- Create/Edit Farmer & Crop records
- Update crop status & growth stages
- View all farm data across operations
- Generate reports & export data

Responsible: Daily Farm Operations

Future Scope

3. Farmer / End User (Future Scope)

- View own crop details & records
- Limited access to assigned farms only
- View dashboards & summaries

Planned: Self-Service Portal Access

Permissions Matrix Summary:

ACTION	System Administrator	Farm Manager	Farmer / End User
1. Manage Configuration ✓	-	-	-
2. Manage Crop/Farmer Data ✓	✓	-	-
3. View Reports/Dashboards ✓	✓	✓	✓

Designed for scalability • Role Hierarchy enforced via Salesforce Profiles & Permission Sets • Future scope for Farmer Portal access via Experience Cloud

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SECURITY REQUIREMENTS

Ensuring data security, compliance, and trusted access for Farm Management System on Salesforce Platform

1. Authentication

- Multi-Factor Authentication (MFA)
- Login Policies

2. Authorization

- Profiles: Sys Admin, Farm Manager, Custom Profiles
- Roles & Role Hierarchy
- Permission Sets
- OWD: Private / Public Read Only
- Sharing Rules

3. Data Security

- Validation Rules for Phone & Aadhaar
- Field-Level Security (FLS)
- Encryption (Platform Encryption / Shield)

4. Monitoring

- Audit Trail
- Login History
- Debug Logs

Best Practices: Regular reviews • Principle of Least Privilege • Compliance & Data Privacy